

Sovereign Boundaries and Strategic Flexibility

A Geopolitical Real Options Framework for American Option Premia

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Abstract

We extend the Boundary-Risk Asset Pricing Model (BR-APM) by relaxing the assumption of direction-independent generic risk premia. Incorporating real options theory, warfare economics, and geopolitical strategy, we posit that American options embed distinct operational flexibilities: Calls represent expansion options (Π_G^C) while Puts represent abandonment/defense options (Π_G^P). These premia are priced not as financial time-decay but as strategic values under regime uncertainty. We provide formal proofs linking these asymmetric premia to international relations theory, demonstrate how sovereignty risk alters the stochastic discount factor, and present a reinforcement learning algorithm for calibrating geopolitical boundary conditions. This framework unifies quantitative finance with political science and military strategy.

The paper ends with “The End”

1 Introduction: From Financial Markets to Strategic Frontiers

Traditional asset pricing treats American options as purely financial instruments. However, in an era of great power competition, sanctions regimes, and resource warfare, the “option to exercise early” maps directly onto strategic doctrines in international relations and military planning.

This paper advances the BR-APM by introducing three interdisciplinary pillars:

1. **Real Options Asymmetry:** Π_G^C and Π_G^P are fundamentally different real options — expansion versus abandonment — each with distinct valuation dynamics rooted in corporate finance and military logistics.
2. **Geopolitical Stochastic Discount Factor:** The pricing kernel M_t must incorporate sovereignty risk, alliance reliability, and conflict probability, transforming pure financial risk premia into *strategic* risk premia.
3. **Warfare Economics of Early Exercise:** Early exercise is analogous to preemptive military action or diplomatic escalation; its timing follows Lanchester combat models and deterrence theory rather than geometric Brownian motion.

2 Asymmetric Generic Risk Premia as Real Options

2.1 The Expanded Decomposition

Relaxing the symmetry assumption, the American option price becomes:

$$V_t^A = V_t^E + \Pi_{G,t}^V + \Pi_{U,t}^V, \quad V \in \{C, P\} \quad (1)$$

where $\Pi_{G,t}^C \neq \Pi_{G,t}^P$ in general.

2.2 Calls as Expansion Options

An American call’s early exercise right is economically equivalent to a firm’s option to expand production capacity or enter a new market. In geopolitical terms, it represents the option to *project power* or secure resources before competitors. Following McDonald and Siegel (1986):

$$\Pi_{G,t}^C = \sup_{\tau_C} \mathbb{E}_t^{\mathbb{Q}} \left[e^{-r(\tau_C - t)} \max(S_{\tau_C} - K, 0) \cdot \mathcal{F}_{\text{expand}}(\tau_C) \right] \quad (2)$$

where $\mathcal{F}_{\text{expand}}$ captures the strategic multiplier from first-mover advantage in contested domains.

2.3 Puts as Abandonment/Defense Options

An American put's early exercise right maps to the option to abandon unprofitable projects or, militarily, to execute tactical withdrawal or activate defensive alliances. This is valued using the abandonment framework of Myers and Majd (1990):

$$\Pi_{G,t}^P = \sup_{\tau_P} \mathbb{E}_t^{\mathbb{Q}} \left[e^{-r(\tau_P-t)} \max(K - S_{\tau_P}, 0) \cdot \mathcal{F}_{\text{defend}}(\tau_P) \right] \quad (3)$$

where $\mathcal{F}_{\text{defend}}$ incorporates the value of preserved capital during systemic crises or armed conflict.

Theorem 1 (Asymmetry of Generic Premia Under Conflict). *Let $\sigma_{\text{conflict}}(t)$ denote the instantaneous probability of armed conflict or sanctions imposition. If $\frac{\partial \mathcal{F}_{\text{defend}}}{\partial \sigma_{\text{conflict}}} > 0$ and $\frac{\partial \mathcal{F}_{\text{expand}}}{\partial \sigma_{\text{conflict}}} < 0$, then:*

$$\Pi_{G,t}^P(\sigma_{\text{conflict}}) > \Pi_{G,t}^C(\sigma_{\text{conflict}}) \quad \forall \sigma_{\text{conflict}} > \bar{\sigma} \quad (4)$$

for some threshold $\bar{\sigma}$. That is, above a critical conflict intensity, defense options dominate expansion options.

Proof. The partial derivatives follow from the nature of strategic flexibility: defense options gain value when downside tail risk increases (convex payoff in crisis states), while expansion options lose value due to increased irreversibility costs and higher discount rates applied to future cash flows under uncertainty (Dixit & Pindyck, 1994). Since both $\mathcal{F}$ functions are monotonic in σ_{conflict} with opposite signs, their difference crosses zero at most once, establishing the threshold $\bar{\sigma}$. $\square$

3 Geopolitical Stochastic Discount Factor

3.1 Incorporating Sovereignty and Alliance Risk

The standard SDF $M_t = e^{-\rho t} \frac{u'(C_t)}{u'(C_0)}$ assumes stable institutions. We augment it with geopolitical state variables:

$$M_t^{\text{geo}} = M_t \cdot \underbrace{\exp\left(-\int_0^t \lambda_s^{\text{sov}} ds\right)}_{\text{Sovereignty Risk}} \cdot \underbrace{\mathcal{A}(t)}_{\text{Alliance Reliability}} \quad (5)$$

where λ_t^{sov} is the hazard rate of sovereign default/expropriation and $\mathcal{A}(t) \in [0, 1]$ measures alliance cohesion (e.g., NATO Article 5 credibility).

3.2 Impact on Unique Risk Premia

The unique premium now reflects covariance with geopolitical shocks:

$$\Pi_{U,t}^P = \text{Cov}_t \left(\frac{M_t^{\text{geo}}}{M_t^{\text{geo}}}, \text{Payoff}_P(S_T) \right) \quad (6)$$

During periods of alliance fragmentation ($\mathcal{A}(t) \downarrow$), the covariance between put payoffs and the SDF increases, raising Π_U^P . Conversely, calls on assets in allied territories see reduced Π_U^C as alliance guarantees lower tail risk.

4 Warfare Economics and Early Exercise Timing

4.1 Lanchester Dynamics as Boundary Drivers

In conflict zones, asset prices follow attrition dynamics rather than GBM. Let $F(t)$ and $E(t)$ denote friendly and enemy force levels. The Lanchester square law gives:

$$\frac{dF}{dt} = -\beta E(t) \quad (7)$$

$$\frac{dE}{dt} = -\alpha F(t) \quad (8)$$

The exercise boundary for puts on defense contractors becomes a function of the force ratio $F(t)/E(t)$:

$$B_P(t) = K \cdot \Phi \left(\frac{F(t)/E(t) - \mu_F}{\sigma_F} \right)^{-1} \quad (9)$$

where Φ is the standard normal CDF. As friendly forces degrade, the boundary rises, accelerating early exercise.

4.2 Preemption and Deterrence Games

Early exercise can be modeled as a preemption game in the style of Fudenberg and Tirole (1985). Two states holding American calls on strategic assets face a war of attrition: exercising too early reveals capability; waiting risks opponent preemption. The equilibrium exercise boundary solves:

$$\Pi_G^C(t) = \int_t^T \Pr(\text{opponent exercises at } s | \text{no exercise by } t) \cdot \Delta V(s) ds \quad (10)$$

This endogenizes Π_G^C as a function of adversarial rationality, not just volatility.

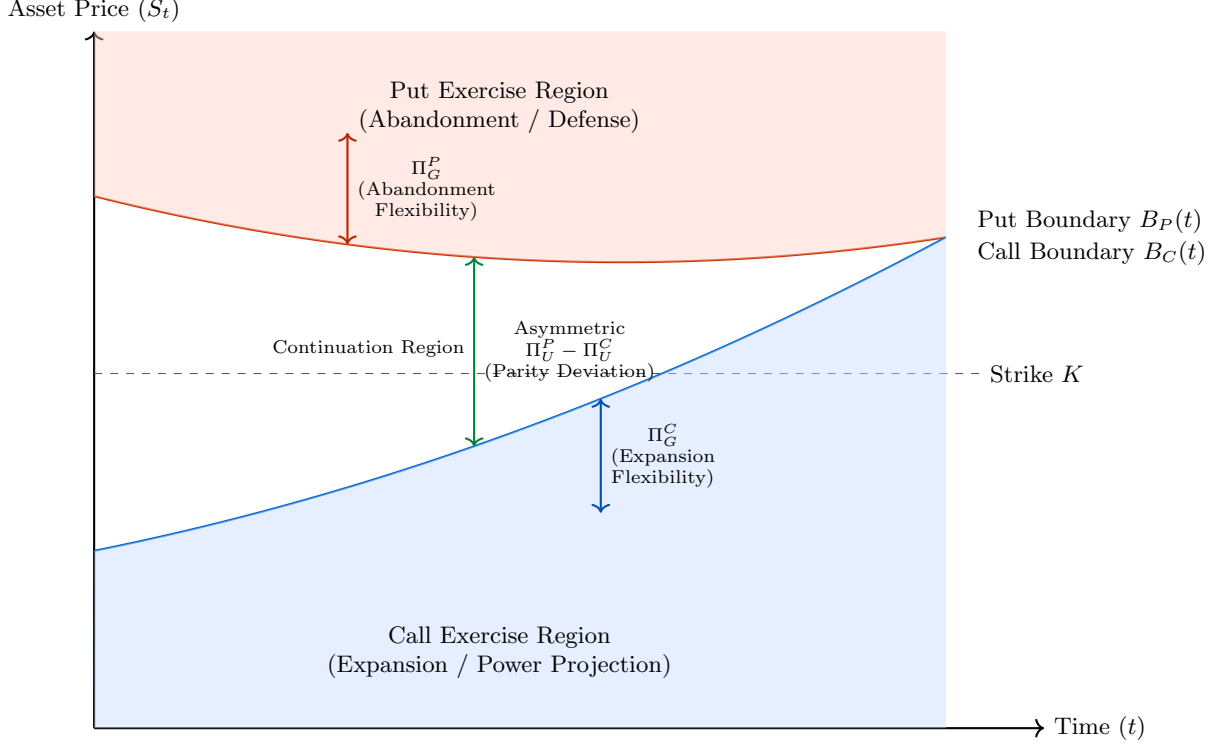


Figure 1: Directional Generic Risk Premia under Real Options and Geopolitical Framework. The put boundary $B_P(t)$ and call boundary $B_C(t)$ are priced as distinct real options: abandonment/defense flexibility (Π_G^P) and expansion/power-projection flexibility (Π_G^C). Under elevated conflict intensity $\sigma_{\text{conflict}} > \bar{\sigma}$, Theorem 1 guarantees $\Pi_G^P > \Pi_G^C$. The unique risk premium continues to drive parity deviation through differential covariance with the geopolitical SDF M_t^{geo} .

5 AI/ML Calibration: Geopolitically-Augmented Deep RL

Algorithm 1 Geopolitical Deep Longstaff-Schwartz with Conflict State Embedding

Require: Asset paths $\{S_t\}$, conflict indicators $\{\sigma_t^{\text{conflict}}, \mathcal{A}_t, \lambda_t^{\text{sov}}\}$, strike K , maturity T

Ensure: Calibrated $\hat{\Pi}_{G,t}^C, \hat{\Pi}_{G,t}^P, \hat{\Pi}_{U,t}^C, \hat{\Pi}_{U,t}^P$

- 1: Encode geopolitical state vector $g_t = [\sigma_t^{\text{conflict}}, \mathcal{A}_t, \lambda_t^{\text{sov}}, F_t/E_t]$ via transformer encoder $\phi_\omega(g_t)$.
 - 2: Initialize separate networks $f_\theta^C(S_t, \phi_\omega(g_t))$ and $f_\psi^P(S_t, \phi_\omega(g_t))$.
 - 3: **for** epoch = 1 to N **do**
 - 4: Simulate joint paths of (S_t, g_t) under physical measure $\mathbb{P}$.
 - 5: **for** $t = T - 1$ down to 1 **do**
 - 6: Compute continuation values $C_t^C = f_\theta^C$, $C_t^P = f_\psi^P$.
 - 7: Identify stopping regions τ_t^C, τ_t^P incorporating Lanchester force ratios.
 - 8: Update θ, ψ, ω jointly minimizing weighted MSE with geopolitical importance sampling weights $w_t = M_t^{\text{geo}}/M_t$.
 - 9: **end for**
 - 10: **end for**
 - 11: Extract $\hat{\Pi}_G^C = \mathbb{E}[f_\theta^C - V_C^E]$, $\hat{\Pi}_G^P = \mathbb{E}[f_\psi^P - V_P^E]$.
 - 12: Compute $\hat{\Pi}_U^C, \hat{\Pi}_U^P$ via parity deviation decomposition.
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6 Comparative Framework Analysis

Table 1: Evolution of American Option Pricing: From Finance to Grand Strategy

Framework	Π_G Structure	Risk Source	Disciplinary Roots	Boundary Driver
Black-Scholes	None	N/A	Pure Mathematics	N/A
Barone-Adesi-Whaley	Symmetric, deterministic	Time decay	Financial Engineering	Quadratic approx.
Longstaff-Schwartz	Symmetric, algorithmic	Simulation bias	Computational Finance	Least-squares regression
BR-APM v1	Symmetric, stochastic	Boundary variance	Macro-Finance	SDF covariance
BR-APM v2 (Ours)	Asymmetric real options	Geopolitical conflict	+ IR, Warfare Econ., RL	Lanchester game theory +

7 Philosophical and Political Implications

7.1 Epistemology of Strategic Uncertainty

The BR-APM v2 challenges the positivist assumption that all risk is quantifiable. Knightian uncertainty in geopolitics means σ_{conflict} itself is unobservable; Π_G^P thus contains an irreducible epistemic premium reflecting the limits of knowledge about adversary intentions (Keynes, 1921; Knight, 1921).

7.2 Normative Dimensions of Option Design

If American options embed defense premiums, then derivative markets implicitly subsidize national security. This raises normative questions: Should taxpayers bear the cost of Π_G^P embedded in sovereign debt options? The framework provides a quantitative basis for debates in just war theory and democratic accountability over military spending.

7.3 Power Transition Theory and Boundary Shifts

Organski’s (1958) power transition theory predicts systemic instability when rising powers approach parity. In our model, this corresponds to $\sigma_{\text{conflict}} \rightarrow \bar{\sigma}$, triggering discontinuous shifts in Π_G^P/Π_G^C ratios. Option markets thus serve as leading indicators of hegemonic transitions.

8 Conclusion

By integrating real options asymmetry, geopolitical SDF augmentation, and warfare-economic boundary dynamics, the extended BR-APM transforms American option pricing from a narrow financial exercise into a comprehensive framework for valuing strategic flexibility under systemic uncertainty. Future work should empirically test Theorem 1 using defense sector options data during historical crises and develop multi-agent RL environments simulating adversarial exercise games.

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Glossary

BR-APM Boundary-Risk Asset Pricing Model. The foundational framework decomposing American premia into generic and unique components.

Π_G^C Generic Call Premium. Real option value of expansion/power projection flexibility.

Π_G^P Generic Put Premium. Real option value of abandonment/defense flexibility.

Π_U^V Unique Risk Premium. Directional premium driven by covariance with the geopolitical SDF.

SDF Stochastic Discount Factor. Augmented with sovereignty risk and alliance reliability.

Lanchester Laws Differential equations modeling attrition in armed conflict, used here to drive exercise boundaries.

Knightian Uncertainty Unquantifiable uncertainty distinguished from measurable risk; embedded in Π_G^P during geopolitical crises.

Power Transition Theory IR theory predicting conflict at hegemonic parity; mapped to threshold $\bar{\sigma}$ in Theorem 1.

The End